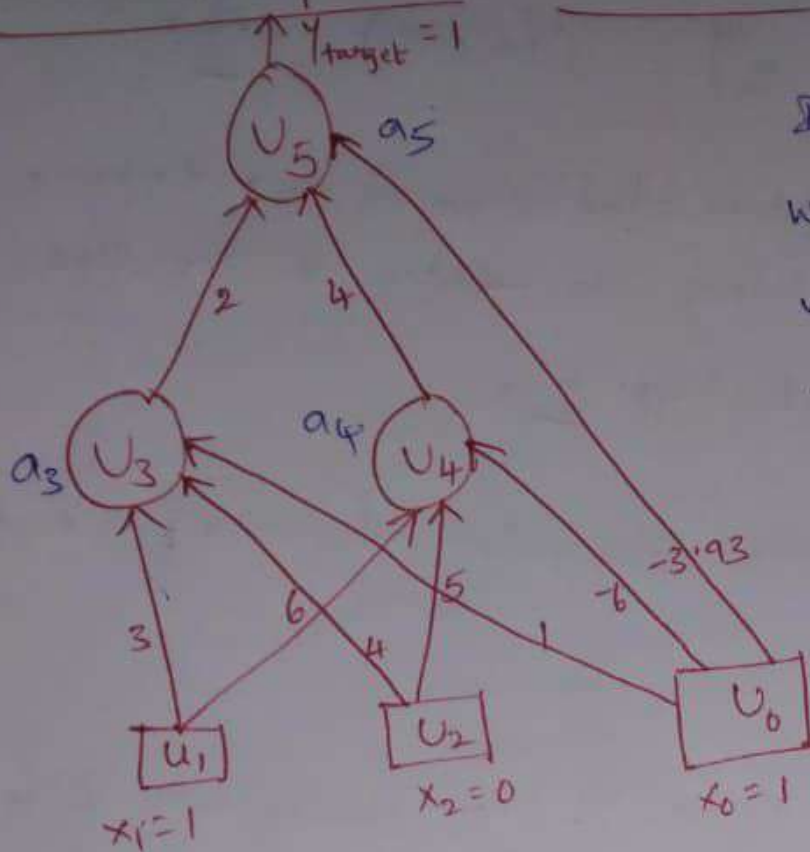


1 hidden unit.

Numerical Problem on Back propagation

①



In the given problem.

$$w_{13} = 3, w_{35} = 2$$

$$w_{24} = 5, w_{45} = 4$$

$$w_{14} = 6, w_{23} = 4$$

also from bias:

$$w_{04} = -6$$

$$w_{05} = -3.93$$

Inputs given are $x_1=1, x_2=0, x_0=1$

lets solve the problem by forward passing:

$$a_j = \sum_i w_{ij} x_i \quad y_j = f(a_j) = \frac{1}{1+e^{-a_j}}$$

$$a_3 = (1 \times 3) + (0 \times 4) + (1 \times 1) = 3 + 0 + 1 = 4$$

$$y_3 = \frac{1}{1+e^{-4}} = \frac{1}{1+0.0183} = \frac{1}{1.0183} = 0.982$$

$$a_4 = (1 \times 6) + (0 \times 5) + (1 \times -6) = 6 + 0 - 6 = 0$$

$$y_4 = \frac{1}{1+e^{-0}} = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

$$a_5 = (0.982 \times 2) + (0.5 \times 4) + (1 \times -3.93)$$

$$= 1.964 + 2 - 3.93 = 0.034$$

$$y_5 = \frac{1}{1+e^{-0.034}} = \frac{1}{1.9607} = 0.51$$

How may we calculate for hidden units:

$$\delta_4 = y_4 (1 - y_4) \cdot w_{45} \delta_5$$

$$= 0.5 (1 - 0.5) \times (4 \times 0.1225)$$

$$= \boxed{0.1225}$$

(2)

$$\delta_3 = y_3 (1 - y_3) w_{35} \delta_5$$

$$= 0.982 (1 - 0.982) 0.1225 \times 2$$

$$= \boxed{0.0043}$$

Now we set learning rate as $\eta = 0.1$ & update the weights by:

$$\Delta w_{ij} = \eta \delta_j y_i$$

$$\Delta w_{03} = \eta \delta_3 x_0$$

$$= 0.1 \times 0.0043 \times 1 = \underline{0.0004}$$

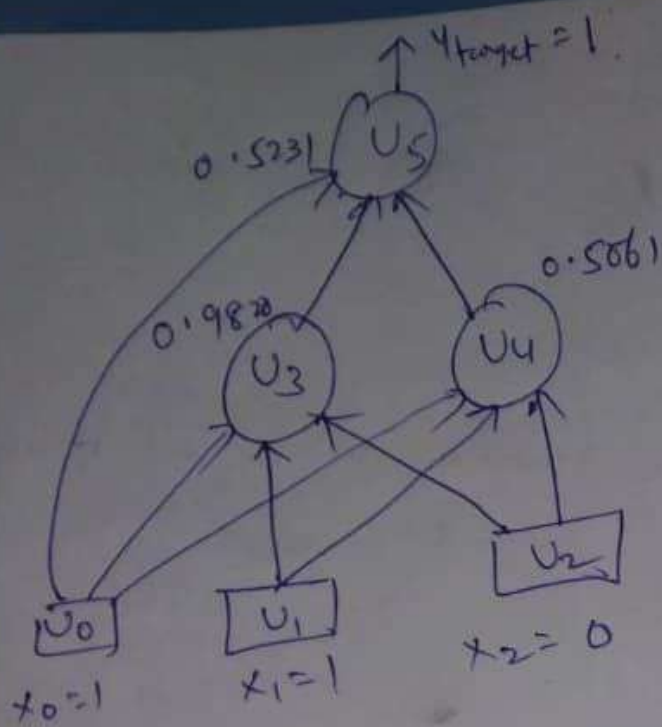
$$\text{So new } w_{03} = w_{03}(\text{old}) + \Delta w_{03}$$

$$= 1 + 0.0004 = \underline{1.0004}$$

$$\text{like wise } w_{13} = \eta \delta_3 x_1 = 0.1 \times 0.0043 \times 3 = 0.0004$$

$$= w_{13}(\text{old}) + \Delta w_{13} = 3 + 0.0004 = \underline{3.0004}$$

Go, change the world



(3)

new forward pass is given by:

$$\begin{aligned}
 a_3 &= \sum_i w_{ij} x_i \\
 &= 1 \times 1.004 + 1 \times 3.004 + 0 \times 4 \\
 &= 1.004 + 3.004 \\
 &= 4.008
 \end{aligned}$$

$$\frac{1}{1 + e^{-4.008}}$$

$$= \frac{1}{1 + 0.0183} = 0.9820$$

$$= \frac{1}{1.0183} = 0.9820$$

$$\begin{aligned}
 a_4 &= 1 \times 5.987 + 1 \times 6.013 + 0 \times 5 \\
 &= -5.987 + 6.013 \\
 &= 0.025
 \end{aligned}$$

$$\frac{1}{1 + e^{-0.025}} = 0.5061$$

$$\begin{aligned}
 a_5 &= 1 \times -3.9078 + 0.9820 \times 4.008 + 0.5061 \times 0.025 \\
 &= -3.9078 + 1.9757 + 0.0126 \\
 &= 0.0799
 \end{aligned}$$

$$\frac{1}{1 + e^{-0.0799}} = \frac{1}{1 + 0.908}$$

$$= 0.524$$

$$\begin{aligned}
 &= -3.9078 + 1.9757 + 0.0126 \\
 &= 0.0799
 \end{aligned}$$

- Time to calculate error:

$$(y_{\text{target}} - y_5) = (1 - 0.51) = \underline{0.49}$$

so we need to reduce the error as our training examples doesn't belong to neither of the class.
as threshold if I keep 0.5.

now let's use back propagation algorithm & find the delta values:

o/p at the y_5 is

$$\delta_k = (t_k - a_k) a_k (1 - a_k)$$

$\therefore$ dealing with 5th neuron.

$$\delta_5 = (t_5 - a_5) a_5 (1 - a_5)$$

(y_{target}) $\uparrow$ y_5 $\downarrow$ obtained

$$= \cancel{(1 - 0.51)} \times \Rightarrow$$

$$= (1 - 0.51) 0.51 (1 - 0.51)$$

$$= 0.49 \times 0.51 \times 0.49$$

$$= \underline{0.1225}$$

$$\Delta W_{23} = \eta \delta_3 \cdot x_2$$

$$= 0.1 \times 0.001 \times 0 = 0$$

$$= \boxed{0}$$

$$\Delta W_{23}(\text{new}) = \Delta W_{23}(\text{old}) + \Delta W_{23}$$

$$= 4 + 0 = \boxed{4}$$

$$\frac{0.4}{W_{04}} = \eta \delta_4 \cdot x_0 = 0.1 \times 0.1225 \times 1$$

$$= 0.01225$$

$$W_{04}(\text{new}) = \Delta W_{04}(\text{old}) + \Delta W_{04}$$

$$= -6 + 0.01225 = \boxed{-5.9878}$$

$$\frac{1.4}{W_{14}} =$$

$$\eta \delta_4 \cdot x_1 = 0.1 \times 0.1225 \times 1$$

$$= 0.01225$$

$$= 6 + 0.01225 =$$

$$\boxed{6.0123}$$

$$\frac{2.4}{W_{24}} =$$

$$0.1 \times 0.1225 \times 2$$

$$= 0$$

$$= \Delta W_{24}(\text{old}) + 0$$

$$\boxed{5}$$

$$\frac{0.5}{W_{05}} =$$

$$\eta \delta_5 \cdot x_0$$

$$0.1 \times 0.1225 \times 1$$

$$= 0.01225$$

$$= -3.93 + 0.01225$$

$$= -3.91$$

$$\frac{3.5}{W_{15}} =$$

$$0.1 \times 0.1225 \times 3$$

$$0.9820$$

$$0.01225$$

$$= 2 + 0.01225$$

$$= 2.01225$$

$$= 2.0120$$

$$\frac{4.5}{W_{25}} =$$

$$\eta \delta_4 \cdot x_2$$

$$0.1 \times 0.1225 \times 0.5$$

$$= 0.006125$$

$$W_{04}(\text{old}) + 0.006125$$

$$= 4 + 0.0061 = 4.0061$$

$$\text{New error} = y_t - y_s$$

$$= 1 - 0.5239$$

$$= 0.476$$

$\therefore$ Error from previous iteration which was 0.49
has been reduced to 0.476

$$(0.490 - 0.476) =$$

$$= 0.014 \quad \text{amount of error has been reduced}$$

X ..